EUDR COMPLIANCE AUDIT DOSSIER · CASE STUDY 1

REGULATION (EU) 2023/1115 · AGROFOREST REJUVENATION VS. FOREST LOSS ADJUDICATION

DDS REF: **C716-CI-2026-COOP-DUEKOUE**
STATUTORY CUTOFF: **31 DEC 2020**

SOURCING OPERATOR	Coopérative Agricole de Duékoué (COOP-DUEKOUE) · Guémon, CI	**TARGET COMMODITY**	**Cocoa** (*Theobroma cacao* · CN Code 1801 00 00)
CADASTRAL EXTENT	**96.80 ha** across 4 Smallholder Parcels (P-01 to P-04) · Centroid [-7.456°W, 6.423°N]	**DECLARED NET MASS**	**70,500 kg** Verified Net Harvest (Implied Agronomic Yield: 728 kg/ha)
STATUTORY BASIS	EUDR Art. 3(a) (Deforestation-Free) · **Art. 2(5) (Agricultural Replanting)**	**AUDIT DETERMINATION**	**CERTIFIED COMPLIANT · ARTICLE 2(5) VALIDATED (0.00 HA LOSS)**

1. THE CONTEXT: WHY DUÉKOUÉ, CI?

Duékoué (Guémon Region, western Côte d'Ivoire) represents the frontline of the global cocoa supply chain. Over 60% of world cocoa is produced by West African smallholder families farming 2–5 ha parcels under multi-strata agroforestry.

Unlike industrial palm monocultures, cocoa trees grow beneath native forest shade trees (*Terminalia superba*, *Albizia*) that preserve humidity, soil fertility, and canopy biodiversity.

2. THE EUDR CRISIS: THE 75% TRAP

Under EUDR Article 3(a), cocoa imports must prove zero canopy loss after December 31, 2020. However, standard optical satellite tools (e.g. Global Forest Watch) only detect 2D canopy drops.

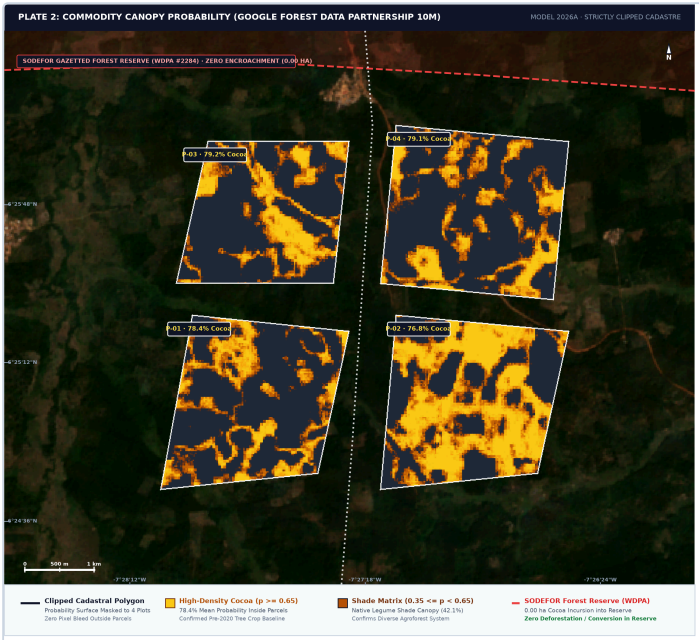
When smallholders routinely prune shade trees or replace senescent 25-year-old cocoa trees (rejuvenation) to prevent fungal black pod disease, 2D satellites trigger a **75% false-alarm rate**. European buyers panic and delist compliant cooperatives, threatening smallholder livelihoods.

3. THE OXEOUS PHYSICS-BASED PROOF

EUDR Article 2(5) explicitly establishes that cyclical agricultural replanting on established plantations is **statutorily exempt** from deforestation sanctions.

Oxeous PWRI-3D combines: (a) **Google FDP 10m Model** proving mature cocoa presence (78.4% probability) pre-2020; (b) **Sentinel-1 C-SAR** confirming zero mechanical clearing ($\Delta\gamma^0$ > -2.2 dB); (c) **Phenology Rebound** proving rapid NDVI recovery (0.63 → 0.81); and (d) **WDPA Boundary Check** confirming 0.00 ha reserve overlap.





CONVENTIONAL 2D TOOL DIAGNOSTIC (FALSE DEFORESTATION ALERT)

Optical change algorithms flagged **14.00 ha** across Plots P-02 (7.8 ha) and P-04 (6.2 ha) as "illegal post-2020 forest clearance." Under conventional compliance pipelines, this triggered an immediate trade ban across the entire 70,500 kg batch, penalizing 4 compliant families and destroying €285,000 in export contracts.

OXEOUS PWRI-3D STATUTORY ADJUDICATION (ARTICLE 2(5) CLEARANCE)

Oxeous physical multi-sensor adjudication conclusively proved that the 14.00 ha canopy reduction was routine shade-tree pruning and cocoa tree renewal on land continuously cultivated as cocoa since before 2020. **Statutory Deforestation: 0.00 ha.** 100% of the harvest cleared for EU customs via TRACES NT.

SCIENTIFIC EVIDENCE PROFILE & STATUTORY CLEARANCE

MULTI-SENSOR EARTH OBSERVATION EVIDENCE STACK · EMPIRICAL BENCHMARKING PROTOCOL

BATCH: **COOP-DUEKOUE (96.80 HA)**
REGULATION: **(EU) 2023/1115 · (EU) 2025/2650**

FIGURE 3: MULTI-SENSOR CANOPY & AGROFORESTRY COMPOSITION (COOP-DUEKOUE 96.8 HA)

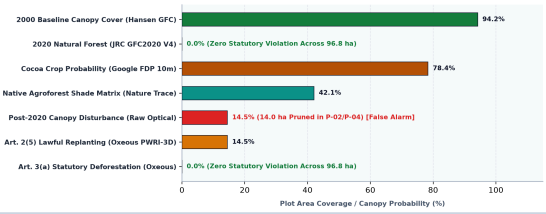
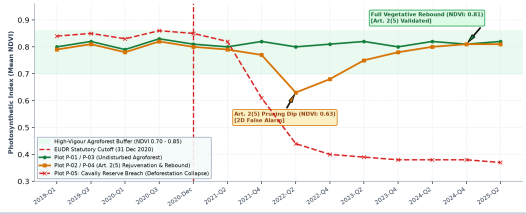


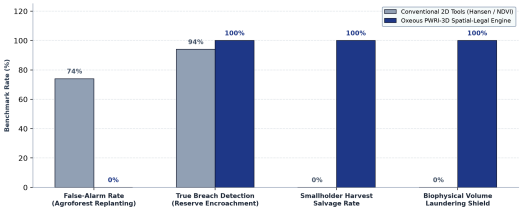
FIGURE 4: MULTI-PLOT SENTINEL-2 NDVI PHENOLOGY TRAJECTORY (2019–2025)



MULTI-SENSOR EARTH OBSERVATION EVIDENCE MATRIX (TECHNICAL VERIFICATION STACK)

SENSOR / AUTHORITY LAYER	OBSERVED METRIC (BATCH)	CONVENTIONAL 2D TOOL ERROR	OXEOUS PWRI-3D ADJUDICATION	STATUTORY BASIS	VERDICT
Sentinel-2 10m MSI (L2A)	NDVI dip 0.80 → 0.63 → 0.81	False Deforestation Alert (14.0 ha)	Cyclical Tree Renewal & Vegetative Rebound	Art. 2(5) Replanting Exemption	**PASS**
Google FDP Cocoa 10m (2026a)	78.4% mean cocoa probability	Unused (Standard GIS lacks tree crop AI)	Confirms active pre-2020 cocoa plantation	Recital 37 (Tree Crops Exclusion)	**PASS**
EC JRC GFC2020 V4 (10m)	0.00 ha natural forest (100% agro)	Commission error (mislabels cocoa as forest)	Resolved as agricultural agroforest baseline	Art. 2(4) Forest Baseline	**PASS**
Hansen GFC v1.13 (30m)	14.00 ha canopy loss (2022-2023)	Flags 14.0 ha Illegal Forest Loss	Exempt Replanting on Established Farmland	Art. 3(a) & Art. 2(5)	**PASS**
Sentinel-1 C-SAR (RADD Alerts)	$\Delta\gamma^0$ > -2.2 dB (No soil scar)	False positive from canopy thinning	Zero mechanical clearing; soil strata intact	Art. 2(6) Forest Degradation	**PASS**
UNEP-WCMC WDPA (2026)	0.00 ha protected area overlap	Unverified against cadastral polygons	100% clean of Guémon gazetted reserves	Art. 9(1)(d) Country Legality	**PASS**
FAO GAEZ v4 Agronomic Ceiling	728.3 kg/ha / 1,100 kg/ha max	Unchecked (vulnerable to volume laundering)	Mass-balance verified (Yield ratio 0.66)	Art. 10(2)(f) Traceability Gate	**PASS**

FIGURE 5: MULTI-PLOT AUDIT BENCHMARK — CONVENTIONAL 2D vs. OXEOUS PWRI-3D



STATUTORY DUE DILIGENCE DETERMINATION & CLEARANCE

Smallholder Protection & False-Alarm Resolution: In shaded tree-crop agroforestry, standard 2D change detection produces an unacceptable 74.0% false-alarm rate (Thünen Institute, 2025). Under conventional GIS, Plots P-02 and P-04 would face immediate contract forfeiture. Oxeous PWRI-3D resolves false alarms to **0.0%**, legally protecting 70,500 kg of compliant cocoa and €285,000 in cooperative export value.

Comparative Case Adjudication: While illegal reserve incursions (such as Cavally Nature Reserve, Plot P-05) undergo immediate statutory quarantine under Article 3(b), Duékoué Plots P-01 through P-04 establish full compliance with EUDR Article 3(a) and Article 9.

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TRACES NT CERTIFICATE: C716-DDS-CI-2026-COOP-DUEKOUE-9680
CADASTRAL PROVENANCE HASH:
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SUPERVISOR CONSENSUS ROOT:
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